

Assignment 2

1. Explain string indexing, slicing, and subscript operator with examples.
2. Write a Python program to reverse a string using slicing.
3. Explain how to convert string to integer and vice versa with examples.
4. Write a program to check whether a string is a palindrome.
5. Explain how text files are handled in Python.
6. Write a program to read and write data to a text file.
7. Write a program to store numbers in a file and calculate their sum.
8. Explain the use of os and sys modules with examples.
9. Write a program to list all files in a directory using os module.
10. Explain list operations (insert, delete, replace) with examples.
11. Write a program to sort a list without using built-in functions.
12. Explain tuples and their immutability.
13. Write a program to search an element in a list.
14. Explain dictionary operations (add, delete, update).
15. Write a program to traverse a dictionary and print keys & values.
16. Define class and object. Write a program using class with attributes and methods.
17. Explain inheritance with example program.
18. Differentiate between method overloading and overriding.
19. Explain data hiding and encapsulation.
20. Explain exception handling in Python. Write a program using
 - try
 - except
 - finally
 - user-defined exception